LABORATORY OBSERVATION

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	2
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1. TITLE

Design Webpage Using HTML5 Canvas & CSS3 Layouts

2. AIM

To construct a responsive styled webpage using HTML5 Canvas drawing, CSS3 flexbox cards, gradient backgrounds, and hover animations.

3. OBJECTIVE

- Implement HTML5 <canvas> element for 2D graphical rendering.
- Draw rectangles, circles, lines, and text using JavaScript Canvas 2D API.
- Design CSS3 flexbox container cards with box-shadows and hover transformations.
- Apply linear gradient background and responsive semantic layout.

4. THEORY

HTML5 Canvas provides a powerful bitmap surface for dynamic, scriptable rendering of 2D shapes and bitmap images using JavaScript. CSS3 provides advanced visual presentation tools including linear gradients, flexbox alignment, smooth transitions, and box-shadow depth.

5. ALGORITHM / PROCEDURE

1. Create index.html with HTML5 semantic header, nav, canvas section, and flexbox cards.
2. Define CSS styling for header, gradient background, flex container, cards, and interactive button hover effects.
3. Access canvas element via document.getElementById('myCanvas').
4. Obtain 2D rendering context using canvas.getContext('2d').
5. Render rectangle with fillRect, circle with arc, line with stroke, and custom text with fillText.
6. Test webpage in browser and verify canvas graphics and card interactions.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Styled Webpage</title>

    <style>
        /* General Styling */
        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #74ebd5, #9face6);
        }

        /* Header */
        header {
            background: #2c3e50;
            color: white;
            text-align: center;
            padding: 20px;
            font-size: 32px;
            text-shadow: 2px 2px 5px rgb(222, 3, 3);
        }
        /* Navigation */
        nav {
            background: #34495e;
            padding: 15px;
            text-align: center;
        }

        nav a {
            color: white;
            text-decoration: none;
            margin: 20px;
            font-size: 18px;
            padding: 8px 15px;
            transition: 0.3s;
            border-radius: 5px;
        }

        nav a:hover {
            background: white;
            color: #34495e;
        }

        /* Canvas Section */
        .canvas-section {
            text-align: center;
            margin: 30px;
        }

        canvas {
            border: 3px solid black;
            background: white;
            box-shadow: 0px 0px 15px gray;
        }

        /* Content Cards */
        .container {
            display: flex;
```

```
            justify-content: center;
            gap: 25px;
            margin: 30px;
            flex-wrap: wrap;
        }

        .card {
            width: 280px;
            background: white;
            border: 2px solid #555;
            border-radius: 15px;
            padding: 20px;
            margin: 10px;
            box-shadow: 5px 5px 15px gray;
            text-align: center;
        }

        .card h2 {
            font-family: Georgia, serif;
            color: darkblue;
            text-shadow: 1px 1px 2px gray;
        }

        .card p {
            font-family: Verdana, sans-serif;
            font-size: 16px;
        }

        /* Button */
        .btn {
            background: #3498db;
            color: white;
            padding: 10px 20px;
            border: none;
            border-radius: 8px;
            cursor: pointer;
            transition: 0.3s;
        }
        .btn:hover {
            background: green;
            transform: scale(1.1);
        }

        /* Footer */
        footer {
            background: #2c3e50;
            color: white;
            text-align: center;
            padding: 15px;
            margin-top: 20px;
        }
    </style>
</head>

<body>

    <!-- Header -->
    <header>
        My Styled Webpage
    </header>

    <!-- Navigation -->
    <nav>
        <a href="#">Home</a>
```

```
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>

    <!-- Canvas -->
    <div class="canvas-section">
        <h2>HTML5 Canvas</h2>

        <canvas id="myCanvas" width="600" height="300">
            Your browser does not support Canvas.
        </canvas>
    </div>

    <!-- Content Cards -->
    <div class="container">

        <div class="card">
            <h2>HTML5</h2>
            <p>
                HTML5 provides the structure of web pages with modern elements
                like canvas, audio, video, and semantic tags.
            </p>
            <button class="btn">Learn More</button>
        </div>

        <div class="card">
            <h2>CSS3</h2>
            <p>
                CSS3 makes websites attractive using gradients, shadows,
                animations, hover effects, and responsive layouts.
            </p>
            <button class="btn">Explore</button>
        </div>

        <div class="card">
            <h2>JavaScript</h2>
            <p>
                JavaScript adds interactivity to web pages and controls HTML5
                Canvas for drawing graphics.
            </p>
            <button class="btn">Read More</button>
        </div>

    </div>

    <!-- Footer -->
    <footer>
        © 2026 My Styled Webpage | Designed by Bhanu Prasad
    </footer>

    <!-- Canvas Script -->
    <script>
        var canvas = document.getElementById("myCanvas");
        var ctx = canvas.getContext("2d");

        // Rectangle
        ctx.fillStyle = "red";
        ctx.fillRect(40, 40, 120, 80);

        // Circle
        ctx.beginPath();
        ctx.arc(280, 80, 50, 0, 2 * Math.PI);
        ctx.fillStyle = "blue";
```

```
        ctx.fill();

        // Line
        ctx.beginPath();
        ctx.moveTo(400, 40);
        ctx.lineTo(550, 150);
        ctx.strokeStyle = "green";
        ctx.lineWidth = 4;
        ctx.stroke();

        // Name Text
        ctx.font = "28px Arial";
        ctx.fillStyle = "black";
        ctx.fillText("Bhanu Prasad", 180, 250);
    </script>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
canvas.getContext('2d')	Retrieves 2D rendering context for drawing API
ctx.fillRect(40, 40, 120, 80)	Draws filled rectangle at specified coordinates
ctx.arc(280, 80, 50, 0, 2*Math.PI)	Draws circle arc in canvas path
ctx.fillText("Bhanu Prasad", ...)	Renders custom student text directly to canvas
display: flex; gap: 20px	Arranges card components in a responsive flex layout
transform: scale(1.1)	Applies smooth scale animation on button hover

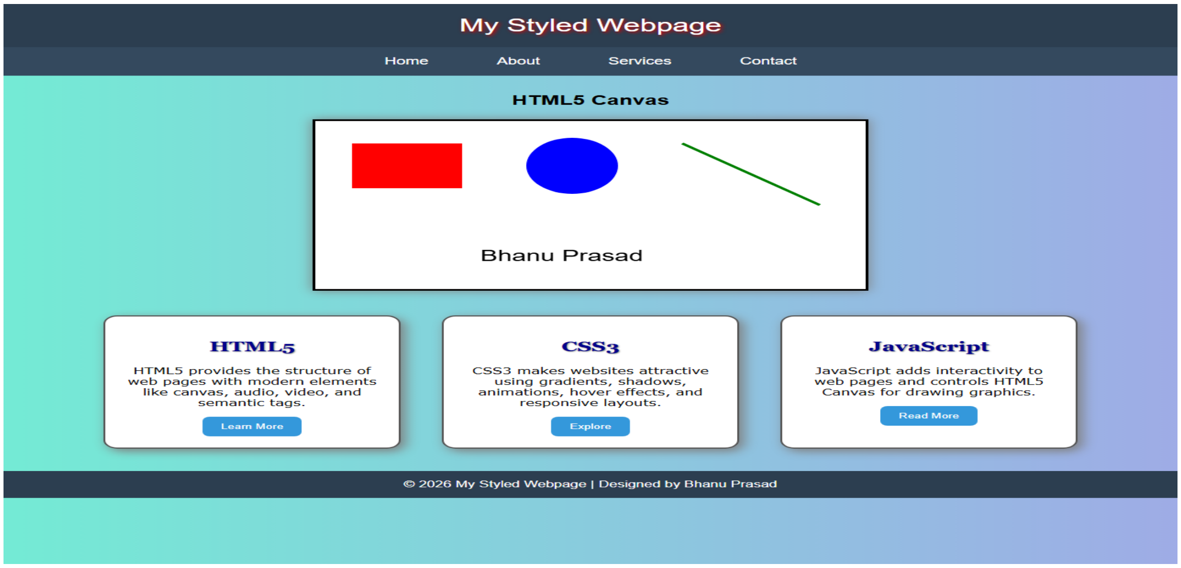
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Rendering	Open index.html in browser	Page renders gradient, header, and cards	**Pass**
TC-02	Canvas Graphics	Check canvas 2D drawings	Rectangle, circle, line, and name text rendered	**Pass**
TC-03	Button Hover	Hover over Learn More button	Button scales up and changes color	**Pass**

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Observation Task (Design Webpage Using HTML5 Canvas & CSS3 Layouts) was successfully executed and verified.